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Author(s): Silvennoinen, Johanna; Kotkajuuri, Jemina; Kujala, Tuomo

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The Effect of Novelty and Typicality on Aesthetic Appeal of Websites

Johanna Silvennoinen, Jemina Kotkajuuri and Tuomo Kujala

Faculty of Information Technology, University of Jyvaskyla, Jyväskylä, Finland

ABSTRACT

More predictability in creating aesthetically appealing websites can be obtained using universal design principles, such as MAYA (Most Advanced, Yet Acceptable). The MAYA principle highlights the importance of both novelty and typicality in aesthetic appeal. While its effect has been empirically demonstrated in various contexts, it has not been examined for websites. Survey participants ($N = 108$) evaluated novelty, typicality, and aesthetic appeal across six commercial and six service websites. The results of a linear mixed-effects model indicate that both novelty and typicality predict website aesthetic appeal, supporting the MAYA principle. However, the connection between novelty and aesthetic appeal is stronger than between typicality and aesthetic appeal. Higher novelty predicts higher aesthetic appeal, particularly for service websites, whereas typicality has a stronger effect on commercial websites. These results explain how the MAYA principle applies to website design and help designers enhance the predictability of creating aesthetically appealing websites.

KEYWORDS

MAYA; novelty; typicality; aesthetic appeal; website

1. Introduction

Aesthetic appeal plays a central role in human–computer interaction (HCI). Due to the inherent subjectivity, complexity, and unpredictability of aesthetic evaluations, extensive research has sought to understand what makes websites visually appealing. As a result, designing aesthetically pleasing websites remains a challenging task. Utilizing universal design principles, more predictability in creating aesthetically appealing websites can be achieved. One such principle is the MAYA (Most Advanced, Yet Acceptable) principle, which highlights the importance of balancing novelty and typicality in aesthetic appeal (Hekkert et al., 2003; Loewy, 1951). Originally introduced by Loewy (1951) and later examined and developed, especially by Hekkert et al. (2003), the MAYA principle is a key component of the unified model of aesthetics (UMA; Hekkert, 2014). UMA aims to explain the experience of aesthetic appreciation in product designs (Berghman & Hekkert, 2017).

The MAYA principle and its underlying mechanisms have been studied across various contexts, including paintings, colors, furniture, music, industrial machinery, and product and service branding. Empirical evidence supports its validity in these domains (Blijlevens, Gemser, et al., 2012; Ceballos et al., 2021; Hekkert et al., 2003; Hung & Chen, 2012; Lidwell et al., 2010; Suhaimi et al., 2022; Thurgood et al., 2014). However, the possible positive effect of the MAYA principle on aesthetic appeal has not been examined in the context of websites. Therefore, currently, there is no empirical evidence of the MAYA principle in the context of websites. In addition, as a result, it remains unclear whether contextual differences influence people's preference for novelty or typicality and how the optimal balance between these variables varies across different artifacts. Additionally, research has produced conflicting findings regarding preferences for typicality and novelty. Given the recurring focus on these variables, the notion of universal mechanisms underlying their interaction emerges. The MAYA

CONTCAT Johanna Silvennoinen johanna.silvennoinen@jyu.fi Faculty of Information Technology, University of Jyvaskyla, Jyväskylä, Finland

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principle, which emphasizes achieving an optimal balance between novelty and typicality, offers a potential explanation for these conflicting results (Lidwell et al., 2010).

The rationale behind the MAYA principle is that designing aesthetically pleasing artifacts requires finding a balance between novelty and typicality (Lidwell et al., 2010). Industrial designer Loewy (1951) proposed that consumers have a threshold for novelty in a product. When this threshold is exceeded, the novelty becomes overwhelming, and the consumer's desire to purchase the product no longer increases—or may even decrease. A skilled designer must recognize when a product's novelty reaches this critical "shock" level. A product that optimally balances novelty and typicality has reached the MAYA level (Loewy, 1951). The validity of the MAYA principle is linked to the psychological tendency of individuals to avoid extremes. This tendency supports the broad applicability of the principle in designing various types of products (Berlyne, 1971; Ceballos et al., 2021).

This study examines whether the MAYA principle explains the aesthetic appeal of websites, whether the balance between novelty and typicality within the MAYA principle varies across commercial and service websites, and whether website type influences the relationship between novelty and aesthetic appeal or between typicality and aesthetic appeal. The results of a linear mixed-effects model indicate that both novelty and typicality predict the aesthetic appeal of websites, providing support for the MAYA principle. Although both factors were statistically significant predictors of aesthetic appeal, the relationship between novelty and aesthetic appeal was stronger than between typicality and aesthetic appeal. The theoretical contribution of the study lies in clarifying the role of the MAYA principle in shaping the aesthetic appeal of websites through empirical evidence. As a practical contribution, the study offers valuable insights for website designers and developers seeking to understand and apply the MAYA principle in designing aesthetically appealing websites.

The rest of the paper is structured as follows. First, previous research on novelty, typicality, and aesthetic appeal is presented. This is followed by the method section, which details the respondents, stimuli, procedure, measurement, and analysis. Next, we present the results and discuss them concerning prior research. We also discuss the limitations of the study. We conclude by summarizing the key findings and outlining directions for future research.

2. Aesthetic appeal, novelty, and typicality

The question of how to create the most aesthetically appealing product is central to design practice, as visual appearance plays a crucial role in how technological products are perceived and appraised. A product's design directly influences its aesthetic appeal and perceived value (Bauerly & Liu, 2008; Moshagen & Thielsch, 2010; Silvennoinen et al., 2014), assessed during the perception process (Ceballos et al., 2021; Creusen & Schoormans, 2005). In diverse design contexts, value can be created by addressing users' aesthetic needs, as consumers increasingly prioritize visual design when selecting products or brands (Bloch et al., 2003; Simonson & Schmitt, 1997). This shift is particularly evident in HCI, where the focus has expanded beyond functionality and usability to acknowledge aesthetic appeal as a central dimension of user experience (Lavie & Tractinsky, 2004; Moshagen & Thielsch, 2013; Udsen & Jørgensen, 2005).

The aesthetic appeal of websites has been studied extensively over the past few decades. Most research has focused on the aesthetic-usability effect (i.e., "*What is beautiful is usable*" Tractinsky et al., 2000), examining the core constituents' roles in shaping visual website experience, their operationalizations, measurements (often psychometric scales), effects on performance, and, for example, moderating variables (e.g., Hassenzahl & Monk, 2010; Thielsch et al., 2019; Thüring & Mahlke, 2007). However, reported results vary due to different conceptualizations and operationalizations of these constituents (e.g., Lima & Gresse von Wangenheim, 2022; Silvennoinen, 2021). Another major line of research pertains to the fastness of first impressions on website aesthetic appeal (e.g., Lindgaard et al., 2006; Skulmowski et al., 2016; Tractinsky et al., 2006; Tuch et al., 2012). In addition, the aesthetic appeal of websites has been examined from the perspective of constructing visual elements on their dynamics in eliciting aesthetic appeal (e.g., Odushegun, 2023; Seckler et al., 2015; Silvennoinen & Jokinen, 2016), also within neuroaesthetics with neuroscientific methods (e.g., Nissen, 2020).

Designers aiming to create highly aesthetic products must carefully consider their approach, given the diverse perspectives and ongoing debates surrounding aesthetics. Hekkert (2006) suggests that the best guidance is to apply universal design principles. These principles, such as the MAYA principle, rooted in the natural ways people perceive and interpret the world, provide a solid foundation for design. The MAYA principle has been shown to explicate the formation of aesthetic appeal in various contexts (e.g., Blijlevens, Carbon, et al., 2012; Blijlevens, Gemser, et al., 2012; Hekkert et al., 2003; Hung & Chen, 2012; Thurgood et al., 2014) but has not yet been examined regarding websites. Thus, the MAYA principle holds potential as an approach for explicating how aesthetic appeal is formed through the balance of novelty and typicality and can be utilized to examine and design for elicited aesthetic appeal of websites. However, to utilize the MAYA principle the researched dynamics between novelty and typicality are needed to be reviewed.

Designers often aspire to introduce technological artifacts that are novel and unprecedented. While this challenges creativity, it is a worthwhile pursuit, as novelty has long captivated people (Radford & Bloch, 2011). Research indicates that novelty is one of the first attributes assessed in a stimulus, with this evaluation occurring at a low level of processing automatically (Ellsworth & Scherer, 2003; Murray et al., 2019; Silvia, 2005). Novelty is also a fundamental factor in aesthetic appeal, making it a crucial consideration in design. The appreciation of novelty has grown in modern society, enhancing its economic value (Kim, 2019). However, despite its recognized importance, research has produced conflicting results about the explanatory power of novelty in shaping aesthetic appeal.

Novelty is generally understood as the originality or newness of a product (Blijlevens et al., 2017). Several studies emphasize its positive effects: people tend to prefer stimuli they perceive as new (Blijlevens, Carbon, et al., 2012; Haig & Whitfield, 2001; Hekkert, 2014), and consumers may deliberately seek atypical product designs (Blijlevens, Carbon, et al., 2012). Novelty shapes first impressions (Kim, 2019), and products perceived as new can stimulate demand, accelerate adoption, and redefine key aspects of consumption (Goode et al., 2013). Psychologically, preference for novelty can be explained by its capacity to offer opportunities for discovery and learning (Hekkert, 2014). Moreover, novelty and exploration possess intrinsic value beyond instrumental benefits (Clark, 2018). At a market level, the consumer population's inherent desire for innovation contributes to its dynamic character (Hirschman, 1980).

Novelty is often understood in relation to typicality (or prototypicality), functioning largely as its opposite (Hekkert et al., 2003; Hung & Chen, 2012). Berlyne (1971) distinguishes between absolute novelty and relative novelty. Absolute novelty refers to cases where a stimulus is entirely new to an individual, whereas relative novelty occurs when a stimulus contains previously encountered elements arranged in a new way. Additionally, Berlyne (1971) differentiates between short-term and long-term novelty, where a stimulus may differ either from experiences within the past few minutes or from all experiences over an extended period. Moreover, novelty has been shown to exhibit a positive linear relationship with trendiness, complexity, and emotion, suggesting that these key dimensions of product semantics can serve as predictors of novelty across domains and sensory modalities (Hung & Chen, 2012).

Research has reported contradictory findings regarding the relationship between novelty and prototypicality, that is, how strongly an object represents its category (Veryzer & Hutchinson, 1998). Some studies indicate that people respond positively to prototypical designs, likely due to a preference for familiarity, the desirable attributes of prototypes, or their informational value. For instance, recent evidence suggests that prototypicality positively influences perceptions of website trustworthiness (Miniukovich & Figl, 2023). Conversely, other studies demonstrate that deviation from prototypicality (i.e., novelty) can enhance aesthetic appeal, either because individuals seek variety or because novel products stand out more distinctly. To reconcile these findings, Veryzer and Hutchinson (1998) proposed an inverted U-shaped relationship between typicality and novelty, whereby people prefer moderate levels of typicality over either extreme typicality or high novelty. This conclusion aligns with Berlyne's (1974) earlier findings, which also identified an inverted U-shaped relationship between novelty and aesthetic preference.

In addition, Hekkert et al. (2003) highlighted conflicting findings on novelty, prompting them to examine the relationship between typicality, novelty, and aesthetic appeal across three studies. Their

hypothesis proposed that both novelty and typicality positively influence aesthetic preference, provided the counter-effect of simultaneous changes in the other variable is controlled. Their findings confirmed that aesthetic appeal in consumer products is shaped by both factors, even for products where aesthetic appeal is not a primary feature. However, they also noted that individual differences, such as expertise, may affect the relative contribution of each factor. Their second study supported these conclusions and confirmed that typicality and novelty jointly determine aesthetic preference, though each suppresses the positive effect of the other. Additionally, experts distinguished more strongly between typicality and novelty than non-experts, but they did not necessarily value novelty more. The third study reinforced these results, providing empirical support for Loewy's (1951) MAYA principle, which suggests that successful design requires balancing novelty and typicality.

Subsequent research has largely validated Hekkert et al.'s (2003) findings (Blijlevens, Gemser, et al., 2012; Diels et al., 2013; Hung & Chen, 2012; Thurgood et al., 2014). Hung and Chen (2012) argued that conflicting findings on preferences for novelty and typicality arise from treating them as opposite ends of a continuum rather than distinct variables, and demonstrated an inverted U-shaped relationship between the two, thereby confirming Hekkert et al.'s (2003) approach. Blijlevens, Carbon, et al. (2012) and Blijlevens, Gemser, et al. (2012) explored context effects on aesthetic appeal, finding that novelty and typicality are distinct, and both significantly influence aesthetic appeal. Thurgood et al. (2014) examined how safety and risk affect product preference for novelty and typicality, confirming that both jointly enhance aesthetic appeal, though they found no evidence that risk manipulation alters this preference.

While most studies support Hekkert et al.'s (2003) findings, some exceptions exist. Diels et al. (2013) tested the MAYA principle in car design using concept art, comparing automotive design students (experts) with psychology students (non-experts). Experts' preferences aligned with the MAYA principle, whereas non-experts favored typicality, supporting prototype preference theory. The authors attributed this divergence to differences in stimuli, study design, and a possible floor effect. Preference for prototypes has also been examined in the context of icon prototypicality, in which higher prototypicality enhances usability (Mao et al., 2025). Beyond expertise, individual traits such as personality and purchase motives influence attitudes toward novelty and typicality (Bloch, 1995; Murray et al., 2019). People who value design are more receptive to new design solutions (Murray et al., 2019), suggesting that different user groups may respond differently to aesthetic innovations. Additionally, context and environment can shape aesthetic preferences.

The MAYA principle shares similarities with Lavie and Tractinsky's (2004) website aesthetics framework, which divides aesthetic perception into classical and expressive aesthetics. Classical aesthetics emphasizes clarity, symmetry, and usability, while expressive aesthetics highlights creativity and originality. Their findings, supported by Kim (2019), indicate that classical aesthetics correlates strongly with perceived usability, while visual novelty aligns with expressive aesthetics. This parallels the MAYA principle, where typicality relates to classical aesthetics and novelty to expressive aesthetics. As an established design principle, the MAYA principle aligns with other universal design principles, such as maximum effect for minimum means and unity in variety (Hekkert, 2006; Post et al., 2016). Designers are encouraged to apply the MAYA principle in mass-market product design, balancing familiarity and innovation to maximize appeal (Lidwell et al., 2010). In addition, recently, a design thinking method combining a multisensory approach and the MAYA principle was developed to aid designers in designing for certain experiences (Silvennoinen & Mononen, 2023).

For the current study, Hekkert's (2014) theory on the MAYA principle serves as the theoretical framework. Research on the MAYA principle is based on empirical aesthetics (e.g., Berlyne, 1974), grounded on the assumption that novelty, typicality, and aesthetic appeal are psychometrically measurable constructs, accessible through empirical methods. This research tradition is based on hypothesis testing, indicating that phenomena can be quantified, hypotheses falsified, and findings generalized. The current study follows this research tradition in examining the MAYA effect in the context of websites.

The MAYA principle operates by context-dependent balance between novelty and typicality, eliciting aesthetic appeal. Thus, according to the MAYA principle, we hypothesize that:

H1: Both novelty and typicality affect aesthetic appeal.

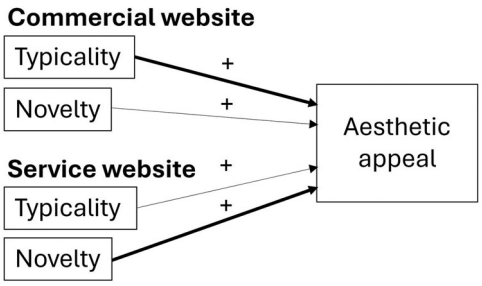


Figure 1. Hypotheses are represented as a conceptual model.

In addition, according to this theory, people seek safety in risky situations, while in safe situations, people seek more novelty. This study examines two types of websites based on Cebi's (2013) classification: commercial websites and service websites. On commercial websites, users purchase products or services, a process Hekkert (2014) associates with a sense of risk due to financial commitment. Thus, based on this premise, the second research hypothesis is:

H2: Typicality is emphasized more in the aesthetic appeal of commercial websites than in the aesthetic appeal of service websites.

In contrast, service websites offer free services and generate revenue through advertisements rather than direct sales. As their use does not involve spending money, service websites may initially evoke a lower sense of risk, in line with Hekkert's (2014) theory. According to this premise, the third hypothesis is as follows:

H3: Novelty is emphasized more in the aesthetic appeal of service websites than in the aesthetic appeal of commercial websites.

To summarize the goals of the current study, the research hypotheses are presented as a conceptual model in Figure 1.

3. Methods

The goal of this study was to examine whether the MAYA principle applies to websites, whether it differs between website types, and whether website type influences the relationship between novelty and aesthetic appeal, as well as typicality and aesthetic appeal. Data were collected through a survey using a within-subjects design.

3.1. Respondents

Respondents ($N = 108$; 47 males, 56 females, 2 non-binary, 3 did not wish to disclose) for the study were recruited via several university mailing lists. The average age of the respondents was 33,6 years ($M = 29$, SD = 13.2, age range 19–82). By education, 2.8% of respondents had completed primary school education, 31.5% had upper secondary education, 43.5% had a bachelor's degree, 20.4% had a master's degree, and 0.9% had doctoral education. The respondents were informed of the ability to withdraw from answering the survey at any point, and all the collected data are anonymized, so no single respondent can be identified. Informed consent was obtained through participants' completion of the survey.

3.2. Stimuli

A total of 12 websites (six service websites and six commercial websites) were selected as stimuli for the study. The selection was based on Cebi's (2013) differentiation of website types. Initially, 25

websites were chosen, and their novelty and typicality were assessed in a preliminary survey. The final 12 websites (Appendix A) were then selected based on the results of this preliminary survey.

In selecting the 25 website landing pages for the preliminary survey, several factors were considered when choosing the stimuli. All the reviewed websites were landing pages of Finnish websites because the effect of the website type was examined and, thus, essential to be communicated to the Finnish respondents of the study. A selection criterion for the commercial websites was that the website presents products or services that are sold on the website. The commerciality was conveyed with pictures on the website showing products for sale or the possibility of renting a service, such as a hotel room. Since the sites were evaluated from an individual's point of view, the stimuli did not include sites from the B2B subcategory, but the websites represented either B2C or C2C sites. Correspondingly, when choosing service websites, the pictures of the website needed to convey information that the website does not sell products or services. The selected sites offer free services as described by Cebi (2013). In the pictures presented on the service websites, there are no elements that indicate a monetary transaction.

The selected 25 website landing pages (12 commercial websites and 13 service websites) were examined to assess the novelty and typicality of the websites (the aesthetic appeal of the websites was not measured here). The purpose of the preliminary survey was to ensure that the stimuli would be selected for the official survey of the study, which comprehensively includes websites located at different points on the scales of high novelty and low novelty and high typicality and low typicality, so that the MAYA principle can be examined more comprehensively.

A total of 14 participants (6 male, 8 female) responded to the preliminary survey, with an average age of 36.5 years (Mdn = 36.5, SD = 11.5, range = 24–57). In the pre-survey analysis, sum variables for novelty and typicality were derived from the statements, and mean novelty and typicality ratings were calculated for each website. Based on these results, 12 commercial and 13 service websites were ranked by their novelty and typicality scores. For the main study, six websites from each category were selected: two representing high novelty/typicality, two representing low novelty/typicality, and two with values near the midpoint. The stimuli (12 website landing pages: 6 commercial, 6 service landing pages varying in novelty and typicality) are detailed in Table A1. In the preliminary survey, Cronbach's alpha values were calculated for the statements measuring novelty and typicality for each website. The high alpha values (>.70) indicated strong internal consistency, providing no reason to question the reliability of the statements at this stage.

3.3. Procedure

The study was conducted as an online survey using the Webropol tool (webropol.fi). The initial section of the survey collected demographic data, including age, gender (male, female, other, prefer not to disclose), and education level. In the next phase, respondents were shown website images one at a time and asked to evaluate them based on predefined statements (see Stimuli section). The order of the website images was randomized.

A within-subjects research design was employed, ensuring that all participants evaluated both commercial and service websites. Compared to a between-subjects design, this approach reduces the risk of group differences affecting reliability (Coolican, 2014). However, a limitation of the within-subjects design is the potential for order effects (Coolican, 2014). To mitigate this, website images were presented in a randomized order, preventing respondents from evaluating all commercial websites before service websites or vice versa.

Respondents were informed in the study invitation that the research focused on evaluating the visual pleasantness of websites. However, they were not explicitly told that the study examined perceived novelty and typicality or that the websites were categorized as service or commercial sites. This omission aimed to prevent influencing participants' evaluations. A within-subjects design increases the likelihood that participants may deduce the study's purpose, potentially leading them to respond in a way they believe aligns with the researcher's expectations (Coolican, 2014). To mitigate this, the exact research focus was withheld to reduce potential bias. Concealing the specific research question did not present ethical concerns that would have made this approach problematic. To uphold ethical research

standards, participant informed consent and anonymity were ensured. The Webropol survey settings guaranteed that no individual participant could be identified from their responses, and all data were processed solely for research purposes.

3.4. Measurement

Before selecting the measurement approach, we reviewed previous studies on the MAYA principle and their methods for assessing key concepts, novelty, typicality, and aesthetic appeal. Research on the MAYA principle frequently employs adjective pairs (Blijlevens, Gemser, et al., 2012; Ceballos et al., 2021; Hekkert et al., 2003; Hung & Chen, 2012) or attitude statements (Berghman & Hekkert, 2017; Thurgood et al., 2014). The measurement scales used in these studies were typically seven-point (Berghman & Hekkert, 2017; Blijlevens, Gemser, et al., 2012; Ceballos et al., 2021; Thurgood et al., 2014) or nine-point (Hekkert et al., 2003; Hung & Chen, 2012). In reviewing prior studies on the MAYA principle, we identified the validated aesthetic pleasure in design (APiD) scale developed by Blijlevens et al. (2017). This scale has been previously utilized in studies such as Berghman and Hekkert (2017) and enables the measurement of typicality, novelty, uniformity, and variability, in addition to aesthetic pleasantness. The APiD scale has been tested in three languages and is generalizable across different cultures (Blijlevens et al., 2017). Within the scale, aesthetic pleasantness and typicality are assessed using five statements, while novelty is measured with four statements. The wording of these statements closely aligns with that used in previous studies on the MAYA principle (see Table 1).

Since participants were required to evaluate 12 websites, special consideration was given to survey length to maintain respondent engagement throughout the study. Utilizing all 14 statements from Blijlevens et al. (2017) scale was deemed impractical, as it could result in an overly lengthy questionnaire. Therefore, three statements were selected for each of the three key concepts (novelty, typicality, and aesthetic pleasantness) to balance measurement reliability with survey efficiency. Blijlevens et al. (2017) note that using only a subset of statements is not problematic. While a single statement could be used in practice, they recommend selecting multiple statements, preferably those with the highest factor loadings or those conceptually most relevant to the given stimulus. Previous studies have also employed partial selections from the scale; for example, Berghman and Hekkert (2017) used three statements to measure aesthetic pleasantness and two statements for both novelty and typicality. The statements selected as operationalizations of the measured variables are presented in Table 1. To mitigate order effects, the measurement items were randomized within the survey.

Results

To ensure the reliability of the selected measurement scales, Cronbach's alphas were calculated for the sum variables assessing novelty, typicality, and aesthetic appeal separately for each website (12). The results indicated acceptable reliability with the Cronbach's alphas (>.70, see Table 1). Furthermore, we conducted confirmatory factor analysis (CFA, RStudio 2025.05.1 + 513) to further validate the reliability of the measurements. For the CFA, the mean of each item across the 12 websites was first calculated for each respondent. The factor loadings in Table 1 indicate that the items align perfectly with the targeted factor structure. Fit indices indicated excellent fit of the targeted factor structure with the data

Table 1. Measurement items of the study, Cronbach's alphas, and factor loadings per item.

Factor	Measurement item	Cronbach's alpha (min of 12)	Factor loading (CFA)
Novelty	"This is a novel website."	.726	.75
	"This design is original."		.92
	"This design is innovative."		.90
Typicality	"This is a typical website."	.853	.92
	"This design is common for a website."		.94
	"This is a standard design."		.89
Aesthetic appeal	"This is a beautiful website."	.857	.92
	"This is an attractive website."		.94
	"This website is pleasing to see."		.91

Table 2. Fit indices for the factor solution, $N = 108$.

Fit indicator	Test statistic	Acceptance criterion	p
χ^2 (24)	28.742	$p > .05$	.230
AIC	863.632	– (relative indicator)	
CFI	0.995	>.95	
TLI	0.992	>.95	
RMSEA	0.043	<.06 ($p > .05$)	.545
SRMR	0.036	<.08	

Table 3. Descriptive statistics.

Variable	Website type	Mean (SD)	Median
Novelty	Commercial	3.53 (1.34)	3.33
	Service	3.21 (1.44)	3.00
Typicality	Commercial	4.68 (1.17)	5.00
	Service	4.61 (1.16)	5.00
Aesthetic appeal	Commercial	4.23 (1.50)	4.33
	Service	3.69 (1.84)	3.33

Table 4. Linear mixed-effects model predicting aesthetic appeal of a website ($N = 1296$).

Fixed effects	Standardized estimate	Estimate	Standard error	p	95 % confidence interval
Intercept	−.055	−1.302	0.199	<.001	−1.693 to −0.912
Type (commercial)	.110	0.187	0.055	.003	0.080 to 0.295
Type (service)	0[a]	0[a]	–	–	–
Novelty	.857	1.034	0.023	<.001	0.989 to 1.079
Typicality	.250	0.363	0.035	<.001	0.294 to 0.432
Random effects		σ^2			
Intercept (participant)		0.095	0.028	<.001	0.053 to 0.170
Intercept (website)		0.615	0.050	<.001	0.524 to 0.722
Residual		0.607			
Intraclass correlation (ICC)					
Participant		0.072			
Website		0.467			
Model fit (−2RLL)		4023.752			

[a]The factor above is compared to the factor with the value of zero.

(Hu & Bentler, 1999; MacCallum et al., 1996; see Table 2). These analyses support the reliability of the shortened and translated APiD scale.

To comprehensively examine the MAYA principle, we aimed to select websites that varied in novelty and typicality. Based on mean values, the websites in the study ranked high in typicality and average in novelty (see Table 3). Aesthetic appeal was rated above average on both website types. However, distributions of the three variables illustrated in Figures B1–B3 indicate good levels of variability in the ratings.

The effects of novelty, typicality, and website type on aesthetic appeal were analyzed using a linear mixed-effects model. The distribution of the aesthetic appeal variable was not normally distributed (Figure B3), and therefore robust covariances were used to handle violations of model assumptions (SPSS v29.0.2.0). There was a strong inverse correlation between the Novelty and Typicality predictors across the websites ($r = -.52$), but the association was sufficiently low to prevent multicollinearity. Before constructing the initial model, intraclass correlations (ICC) were assessed using an intercept-only model, where aesthetic appeal served as the dependent variable and participant and website as random effects. The ICC value for the participant was 1.2 and 53.2% for the website, indicating that ratings for the same website were significantly more similar to each other than ratings for different websites. The high ICC for the website supports the use of the mixed-effects modeling approach for the analysis. ICC for participants was low for the intercept-only model, but participant was included as a random effect too, to control for possible between-participant variability in the aesthetic appeal.

In the final model with main effects only, all the fixed factors, website type, novelty, and typicality, predicted significant aesthetic appeal (see Table 4). Notably, the novelty's standardized estimate (.857) is much larger than the typicality's estimate (.250). The commercial websites were rated significantly more aesthetically appealing than the service websites (effect as standardized estimate: .110). All the fixed factors significantly improved the model fit (−2RLL). Pseudo-R^2 measures for the model (SPSS

Table 5. Linear mixed-effects model with interaction effects for predicting aesthetic appeal of a website ($N = 1296$).

Fixed effects	Standardized estimate	Estimate	Standard error	p	95 % confidence interval
Intercept	.010	−1.131	0.204	<.001	−1.530 to −0.731
Novelty × type (commercial)	.758	0.959	0.030	<.001	0.901 to 1.018
Novelty × type (service)	.943	1.101	0.025	<.001	1.053 to 1.149
Typicality × type (commercial)	.247	0.421	0.036	<.001	0.351 to 0.491
Typicality type (service)	.230	0.281	0.037	<.001	0.209 to 0.353
Random effects		σ^2			
Intercept (participant)		0.096	0.028	<.001	0.054 to 0.170
Intercept (website)		0.600	0.049	<.001	0.511 to 0.706
Residual		0.598			
Intraclass correlation (ICC)					
Participant		0.074			
Website		0.464			
Model fit (−2RLL)		4007.663			

v.29) were very high, .563 for the fixed effects and .799 with both the fixed and random effects included. After the model construction, the residual plots of the final model were inspected. The inspected plots did not indicate any clear deviations from normality or homoscedasticity (see Figures B4 and B5.

Furthermore, we studied the interaction effects of type × novelty and type × typicality on the aesthetic appeal of a website (Table 5). The linear mixed-effects model indicates that the effect of novelty is stronger for the Service websites than for the Commercial websites (difference in the effect size: .185), whereas the effect of typicality on aesthetic appeal is a bit weaker for the Service websites than for the Commercial websites (difference: .017). The marginal pseudo-R^2 for the fixed effects was .568, and the conditional pseudo-R^2 was .801 for the full model, including the random effects too. Again, the residual plots did not indicate any clear deviations from normality or homoscedasticity (Figures B6 and B7).

4. Discussion

The focus of this study was to examine whether the relationship between novelty, typicality, and aesthetic appeal differs across commercial and service websites. The findings indicate that both novelty and typicality are positively associated with aesthetic appeal, suggesting that both factors are considered simultaneously in aesthetic evaluations. These results support the MAYA principle's core idea, that both novelty and typicality play a crucial role in aesthetic perception, and align with previous research on the topic (Blijlevens, Gemser, et al., 2012; Hekkert et al., 2003; Thurgood et al., 2014).

The mixed-effects model analysis confirmed that both novelty and typicality positively predicted aesthetic pleasantness, providing statistical support for the first hypothesis. This finding aligns with previous research on the MAYA principle. In addition to examining the core idea of the MAYA principle, this study aimed to determine whether the relationship between novelty, typicality, and aesthetic pleasantness differs across website types. The second and third hypotheses were also supported; typicality played a greater role in the aesthetic evaluation of commercial websites than service websites, and novelty was more dominant in evaluations of service websites than commercial websites. While the aesthetic appeal of websites has been studied extensively over the past few decades (e.g., Hassenzahl & Monk, 2010; Lima & Gresse von Wangenheim, 2022; Lindgaard et al., 2006; Silvennoinen, 2021; Thielsch et al., 2019; Thüring & Mahlke, 2007; Tractinsky et al., 2006), research has not focused on the role of the MAYA principle as an explicating factor. The findings of the current study explain the formation of the aesthetic appeal of websites with novelty and typicality.

Based on the mixed-effects models, novelty was a stronger predictor of aesthetic appeal than typicality. This suggests that, while both factors contribute to aesthetic judgment, novelty has a greater impact. This result is not unprecedented. Berghman and Hekkert (2017) reported similarly modest effects of typicality in their study on aesthetic appreciation. Although the MAYA principle emphasizes the importance of both novelty and typicality, typicality's effect was significantly lower than that of other factors in their findings. To explain the weaker effect of typicality, Berghman and Hekkert (2017) reference

Biederman and Vessel (2006) theory, which suggests that people prefer novel stimuli when they can still recognize them. Berghman and Hekkert (2017) argue that participants in their study likely recognized the products used as stimuli, particularly since all the products were already available on the market. They suggest that a stronger typicality effect might emerge if more unconventional or exotic products were included in the study.

The findings on typicality in this study likely follow a similar logic. The stimuli consisted of existing websites, designed according to established web design conventions (e.g., Garrett, 2011; Krug, 2014). These conventions, widely adopted, shape user expectations, as interface elements are typically placed in predictable locations (e.g., navigation menus at the top or left side; Krug, 2014). Deviations from these norms are rare, which was reflected in the dataset: websites generally received higher typicality and lower novelty ratings. The results might differ if more unconventional designs were included. This study examined whether novelty and typicality predict aesthetic pleasantness differently across website types. Mixed-effects models indicated that website type also had a significant effect on aesthetic appeal. The evidence suggests that, in general, when controlling for the effects of novelty and typicality, the commercial websites were experienced as more aesthetically appealing than the service websites.

The main framework considered was Hekkert's (2014) theory on how risk and safety influence preferences for novelty or typicality. However, empirical support for this theory remains inconsistent. While it has been used to explain variations in UMA model preferences across product categories (Berghman & Hekkert, 2017), few studies have demonstrated a direct effect of risk and safety (Thurgood et al., 2014). However, the findings of Thurgood et al. (2014) suggest that risk and safety may not shape the balance between novelty and typicality as proposed. Given the limited research on how risk and safety influence preferences for novelty or typicality remains uncertain and warrants further investigation.

The proposed link between risk and financial expenditure formed the basis of the study's hypotheses: commercial websites, where users purchase products and services, were expected to be perceived as riskier than service websites, which primarily offer free services. Consequently, users were hypothesized to prefer typicality on commercial websites and novelty on service websites. In the current study, risk and safety manifested across website types as hypothesized—that is, novelty was preferred in service websites, while typicality dominated in commercial websites. Hekkert (2014) posited that expensive purchases are perceived as riskier, whereas lower-cost purchases involve less risk. Consequently, people may prioritize safety in high-cost purchases and seek novelty in lower-cost ones.

4.1. Limitations and future research

This study has certain limitations, which are examined through validity perspectives, alongside a discussion of its strengths and potential reliability concerns. Ensuring internal validity requires assessing conceptual appropriateness, theoretical alignment, and measurement reliability. A key aspect of validity is that the chosen instrument accurately measures the intended construct. This study employed the APiD scale (Blijlevens et al., 2017), which has been validated by its developers and applied in prior research, including Berghman and Hekkert (2017). Additionally, the scale aligns with previous operationalizations of the MAYA principle, reinforcing its reliability. The primary concern relates to translation accuracy. Although Blijlevens et al. (2017) validated the scale in multiple languages, no official Finnish translation existed, raising the possibility of validity issues. However, Cronbach's alpha values for the sum variables were high, and CFA gave support for the targeted factor structure, indicating strong internal consistency and supporting the accuracy of the Finnish translation. Based on these evaluations, the measurement instrument can be considered reliable, and it is reasonable to assume that novelty, typicality, and aesthetic pleasantness were measured accurately.

This study focused exclusively on the factors of the MAYA principle, novelty, and typicality, but these are not the only determinants of aesthetic pleasantness. In the UMA model of aesthetic appeal (Berghman & Hekkert, 2017; Hekkert, 2014), the MAYA principle represents just one of three levels, meaning this study does not account for other potential influences on aesthetic appeal. The stimulus selection is also crucial for evaluating the study's reliability. The use of real, interactive websites supports ecological validity, but participants assessed sites solely through static images. In reality, users

evaluate websites not just visually but also through functional and interactive elements. The lack of interaction may have affected participants' evaluations. In future studies, the MAYA principle is needed to be examined also in an interactive setting.

Additionally, despite a pre-survey to assess typicality and novelty, the final dataset was skewed, containing more highly typical and low-novelty websites. Ideally, the dataset should have covered the full range of novelty and typicality for a more comprehensive examination of the MAYA principle. Including more atypical and innovative websites might have strengthened the effect of typicality in the results.

This study classified websites based on Cebi's (2013) categorization of commercial and service websites. While the selection followed Cebi's definitions, this is not the only way to classify websites. Therefore, the findings apply specifically to commercial and service websites as Cebi (2013) defined them. This does not rule out the possibility that differences in the MAYA principle could emerge in other website categories. Additionally, this study examined broad website categories, and distinct patterns might have appeared if subcategories had been analyzed. A deeper understanding of how the balance between novelty and typicality shapes the aesthetic appeal of websites could be elaborated by examining more different website types, which would provide valuable insights for website and user experience design. In addition, the participants evaluated websites based on static images, rather than interacting with them directly. A more naturalistic experimental design, where participants actively navigate and use the websites, might better reveal the potential effects of risk and safety.

The representativeness of participants is another key factor for reliability and generalizability. A total of 108 participants completed the survey, with a sample skewed toward young adults, making the findings highly representative of young adults but less generalizable to older generations. A more age-diverse sample may have produced different results. To facilitate data collection, the survey was conducted online using convenience sampling, allowing participants to respond independently of location and without researcher interaction. However, this means that environmental factors, such as distractions or the device used, may have influenced responses. Despite these limitations, all participants voluntarily completed the survey without compensation, likely responding at a convenient time, which may have reduced response-related biases. Future studies could control possible participant fatigue in more detail. In addition, familiarity with website design, cultural factors, cross-cultural comparisons, longitudinal studies, and digital fluency might influence the results obtained and should be examined and controlled in future research.

The MAYA principle has been extensively studied, supporting its validity. However, certain aspects require further investigation. Further research should explore the impact of risk and safety, particularly by focusing on Hekkert's (2014) specific examples. Additionally, future studies should measure users' perceived risk and safety within the research design to determine whether these factors manifest as expected. Kim (2019) highlighted a lack of research on visual novelty in HCI, calling for more studies on user reactions to visual novelty in websites and applications. Similarly, this study's literature review revealed a gap in research on technological artifacts in studies on the MAYA principle. Future research should broaden its scope beyond the MAYA principle to examine other universal design principles in HCI-driven technological solutions.

The current study followed empirical aesthetics (e.g., Berlyne, 1974) in examining the MAYA effect in the context of websites. However, aesthetic appeal, novelty, and typicality in human experience can be studied with multiple different methodological approaches (e.g., phenomenology), revealing different perspectives on the phenomenon.

5. Conclusions

This study examined the impact of novelty and typicality on aesthetic appeal in commercial and service websites. The findings support the MAYA principle, as both novelty and typicality were positively associated with aesthetic appeal. However, novelty was a stronger predictor of aesthetic appeal than typicality, and the MAYA balance differed between the website types; commercial websites are more strongly connected to typicality, and service websites to novelty. Thus, the MAYA principle is a context-dependent phenomenon, and the influence of variable loadings on the effect should be acknowledged in the methodological approach, for example, when utilizing linear mixed-effects models. In addition,

in the current study, CFA was conducted to further validate the factor structure and reliability of the measurements with the shortened and translated APiD scale.

The results explain the formation of the aesthetic appeal of websites with the MAYA principle and, thus, contribute to the vast line of research on examining how the aesthetic appeal of websites operates. In addition, the results offer valuable insights for website designers, particularly in applying the MAYA principle. Effective design should integrate both elements, satisfying users' desire for new experiences while maintaining familiar features that enhance information processing fluency and provide a sense of security (Hekkert, 2014). Designing websites with the MAYA principle, more predictability on the possibility of aesthetic appeal to emerge can be obtained.

Although both novelty and typicality were statistically significant predictors of aesthetic appeal, the relationship between novelty and aesthetic appeal was stronger than that of typicality. This suggests that while both factors contribute to aesthetics, website designers should prioritize incorporating novel elements. Website structures follow established conventions present across most sites. Users generally expect familiar elements in specific locations, which aids navigation (Roth et al., 2010). It is possible that adhering to these conventions inherently satisfies users' need for typicality, while novel and innovative features enhance engagement within familiar frameworks. For websites that deviate from standard structures, the role of typicality may become even more critical. In such cases, designers should consider integrating familiar elements into unconventional designs to help users navigate effectively and feel comfortable within the digital environment, especially if the design context involves commercial actions.

Each website design process is unique, requiring careful consideration of the site's objectives, target audience, and the website owner's goals. There is no single formula for achieving optimal aesthetic appeal. However, given the vast range of design possibilities, universal design principles, validated by extensive research, provide a strong foundation for guiding the process. As Hekkert (2006) argues, unless a designer has a compelling reason to deviate, adhering to universal design principles is the most effective approach. This study supports Hekkert's (2006) perspective, particularly regarding the MAYA principle. When aiming for aesthetically appealing solutions, it is crucial to carefully balance novelty and typicality in the design.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Author contributions

CRediT: **Johanna Silvennoinen**: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Supervision, Writing – original draft, Writing – review & editing; **Jemina Kotkajuuri**: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing; **Tuomo Kujala**: Data curation, Formal analysis, Investigation, Methodology, Validation, Writing – original draft, Writing – review & editing.

ORCID

Johanna Silvennoinen http://orcid.org/0000-0002-0763-0297
Tuomo Kujala http://orcid.org/0000-0001-8222-8540

Data availability statement

Data will be made available upon request.

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About the authors

Johanna Silvennoinen is a Senior Lecturer of Cognitive Science (University of Jyväskylä, Finland) focusing on cognitive and affective processes in HCI, especially on predictability of experience design, aesthetics, deceptive design, and human–AI co-creation. She is the vice-chair of SIGCHI Finland and an executive committee member of Experience Research Society.

Jemina Kotkajuuri holds a master's degree in Cognitive Science from the University of Jyväskylä, with an academic background in Mathematical Information Technology. She currently works as an accessibility specialist, focusing on evaluating and promoting the inclusiveness of digital services.

Tuomo Kujala is an Associate Professor of Cognitive Science at the Faculty of Information Technology at the University of Jyväskylä and an Adjunct Professor of Cognitive Science at the University of Helsinki. His research interests focus on computational modeling of attentional processes, inattention, and multitasking performance.